# ST117 Individual DRAFT Written Report - Phase 2 Task 3 (Woodland vegetation - Butterflies and Moths)

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## a)

```
# 1. Load data
bf <- read.csv("ECN_IB1.csv", stringsAsFactors = FALSE)   # butterflies
mt <- read.csv("ECN_IM1.csv", stringsAsFactors = FALSE)   # moths
vw <- read.csv("ECN_VW1.csv", stringsAsFactors = FALSE)   # woodland vegetation

# 2. Clean data (based on EDA)

# Butterflies
bf$VALUE <- suppressWarnings(as.numeric(bf$VALUE))
bf$DATE  <- as.Date(bf$SDATE, format = "%d-%b-%y")
bf$YEAR  <- year(bf$DATE)

bf_clean <- bf %>%
  filter(!is.na(VALUE), VALUE > 0, !is.na(YEAR))


# Moths
mt$VALUE <- suppressWarnings(as.numeric(mt$VALUE))
mt$DATE  <- as.Date(mt$SDATE, format = "%d-%b-%y")
mt$YEAR  <- year(mt$DATE)

mt_clean <- mt %>%
  filter(!is.na(VALUE), VALUE > 0, !is.na(YEAR))


# Woodland vegetation
vw$YEAR <- vw$SYEAR

vw_clean <- vw %>%
  filter(!is.na(SITECODE), !is.na(YEAR), !is.na(TREE_SPEC))

# 3. Aggregate vegetation
veg_agg <- vw_clean %>%
  group_by(SITECODE, YEAR) %>%
  summarise(
    unique_species = n_distinct(TREE_SPEC),
    .groups = "drop"
  )


# 4. Butterfly aggregation
bf_agg <- bf_clean %>%
  group_by(SITECODE, YEAR) %>%
  summarise(
    total_butterflies = sum(VALUE),
```

```r
    .groups = "drop"
  )

# Merge with vegetation
bf_final <- bf_agg %>%
  left_join(veg_agg, by = c("SITECODE", "YEAR"))

# Print first 3 rows
head(bf_final, 3)
```

```
## # A tibble: 3 x 4
##   SITECODE  YEAR total_butterflies unique_species
##   <chr>    <dbl>             <dbl>          <int>
## 1 T01       1993               302             NA
## 2 T01       1994               428             NA
## 3 T01       1995               635             NA
```

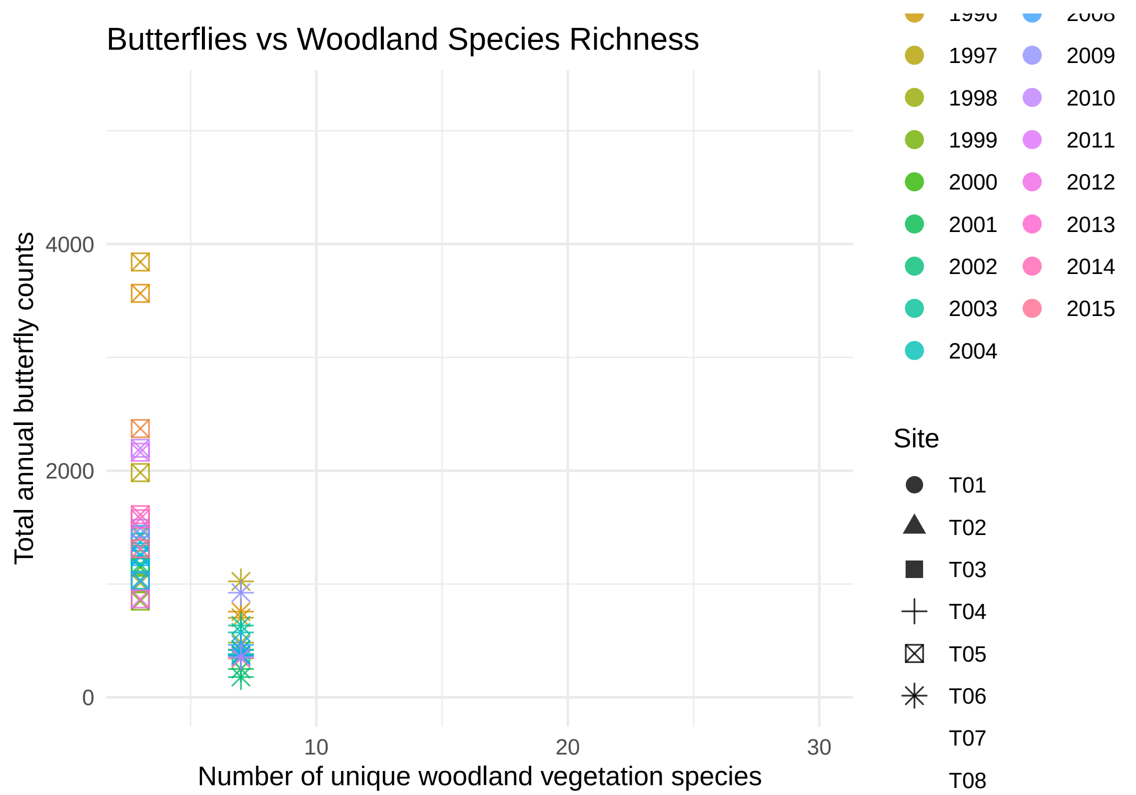
```r
# 5. Moth aggregation
mt_agg <- mt_clean %>%
  group_by(SITECODE, YEAR) %>%
  summarise(
    total_moths = sum(VALUE),
    .groups = "drop"
  )

# Merge with vegetation
mt_final <- mt_agg %>%
  left_join(veg_agg, by = c("SITECODE", "YEAR"))

# Print first 3 rows
head(mt_final, 3)
```

```
## # A tibble: 3 x 4
##   SITECODE  YEAR total_moths unique_species
##   <chr>    <dbl>       <dbl>          <int>
## 1 T01       1993        3141             NA
## 2 T01       1994        3944             NA
## 3 T01       1995        4476             NA
```

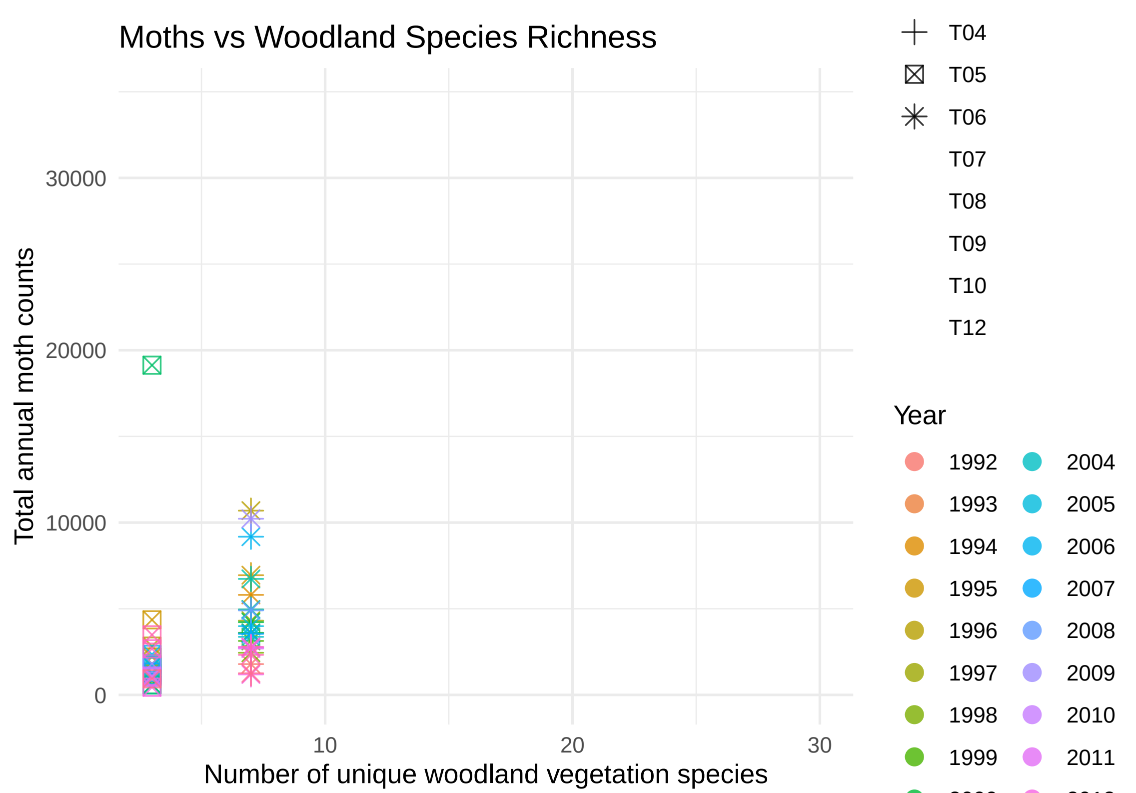
## b)

```r
# 1. Load data
bf <- read.csv("ECN_IB1.csv", stringsAsFactors = FALSE)
mt <- read.csv("ECN_IM1.csv", stringsAsFactors = FALSE)
vw <- read.csv("ECN_VW1.csv", stringsAsFactors = FALSE)

# 2. Clean data

# Butterflies
bf$VALUE <- suppressWarnings(as.numeric(bf$VALUE))
bf$DATE <- as.Date(bf$SDATE, format = "%d-%b-%y")
bf$YEAR <- year(bf$DATE)

bf_clean <- bf %>%
  filter(!is.na(VALUE), VALUE > 0, !is.na(YEAR))

# Moths
mt$VALUE <- suppressWarnings(as.numeric(mt$VALUE))
mt$DATE <- as.Date(mt$SDATE, format = "%d-%b-%y")
mt$YEAR <- year(mt$DATE)
```

```r
mt_clean <- mt %>%
  filter(!is.na(VALUE), VALUE > 0, !is.na(YEAR))

# Vegetation
vw$YEAR <- vw$SYEAR

vw_clean <- vw %>%
  filter(!is.na(SITECODE), !is.na(YEAR), !is.na(TREE_SPEC))


# 3. Aggregate data

# Vegetation richness
veg_agg <- vw_clean %>%
  group_by(SITECODE) %>%
  summarise(unique_species = n_distinct(TREE_SPEC), .groups = "drop")

# Butterflies
bf_agg <- bf_clean %>%
  group_by(SITECODE, YEAR) %>%
  summarise(total_butterflies = sum(VALUE), .groups = "drop") %>%
  left_join(veg_agg, by = "SITECODE")

# Moths
mt_agg <- mt_clean %>%
  group_by(SITECODE, YEAR) %>%
  summarise(total_moths = sum(VALUE), .groups = "drop") %>%
  left_join(veg_agg, by = "SITECODE")


# 4. Scatterplots

# Plot 1: Butterflies
ggplot(bf_agg, aes(x = unique_species, y = total_butterflies,
                   colour = factor(YEAR), shape = SITECODE)) +
  geom_point(size = 3, alpha = 0.8) +
  labs(title = "Butterflies vs Woodland Species Richness",
       x = "Number of unique woodland vegetation species",
       y = "Total annual butterfly counts",
       colour = "Year",
       shape = "Site") +
  theme_minimal()
```




```r
# Plot 2: Moths
ggplot(mt_agg, aes(x = unique_species, y = total_moths,
                   colour = factor(YEAR), shape = SITECODE)) +
  geom_point(size = 3, alpha = 0.8) +
  labs(title = "Moths vs Woodland Species Richness",
       x = "Number of unique woodland vegetation species",
       y = "Total annual moth counts",
       colour = "Year",
       shape = "Site") +
  theme_minimal()
```



## c)

Two clear issues stand out for fitting a simple linear regression.

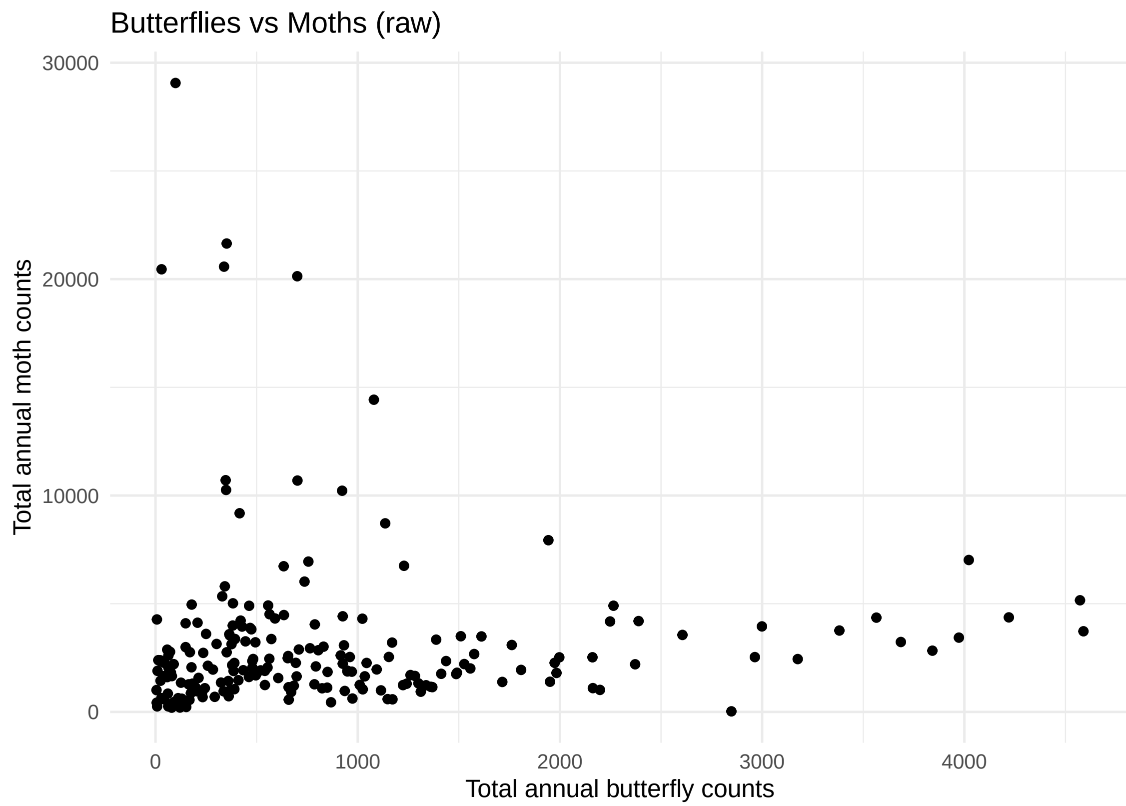
First there is strong heteroscedasticity. The spread of butterfly and moth counts increase species richness changes, especially for lower richness values where counts vary widely. This violates the constant variance assumption of linear regression, meaning the model errors are not evenly distributed. As a result, standard errors and inference from the model would be unreliable.

Second, there is clear clustering and limited variation in the predictor. Most observations are concentrated at just a few values of vegetation richness (around 3 and 7), rather than being spread continuously. This creates effectively grouped data rather than a true continuous relationship, making it difficult for a linear model to capture a meaningful trend. It also suggest potential confounding by site, since richness is largely site-driven, while counts vary strongly by year.

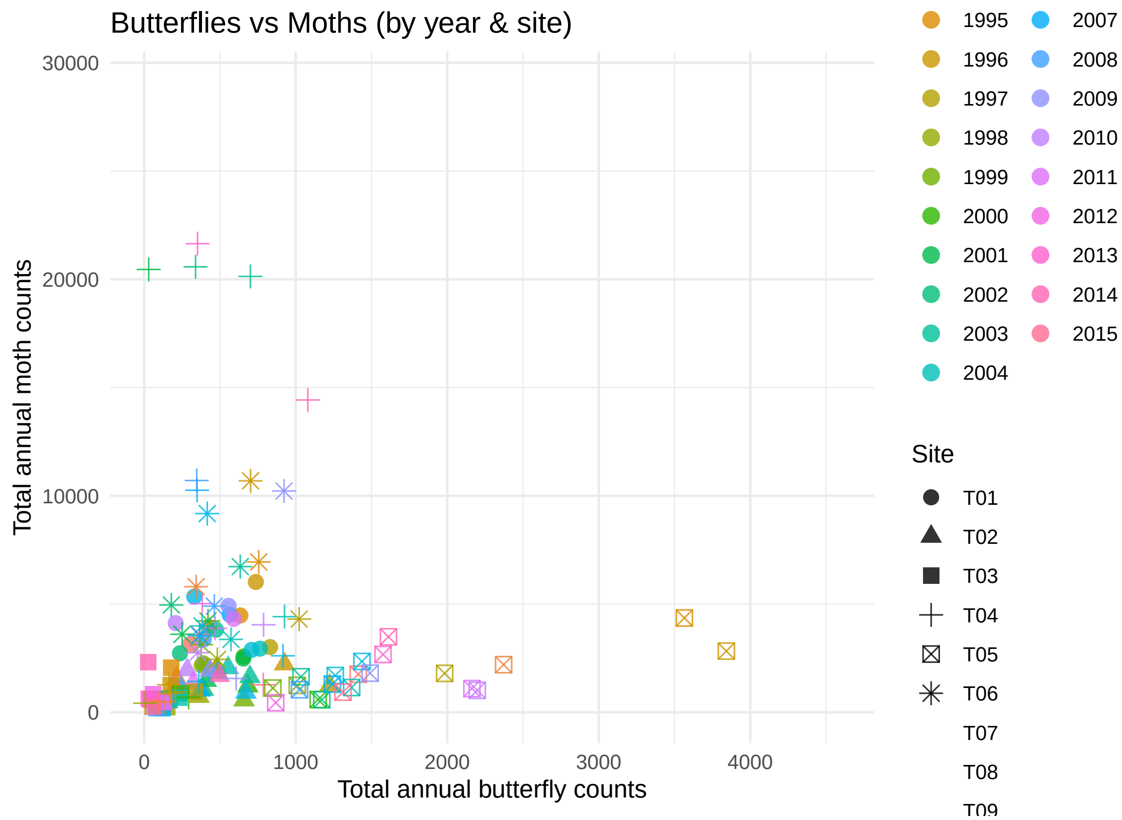
Overall, these issues indicate that a simple linear regression is likely too simplistic, and the relationship may be better explained by site effects or more flexible models.

## d)

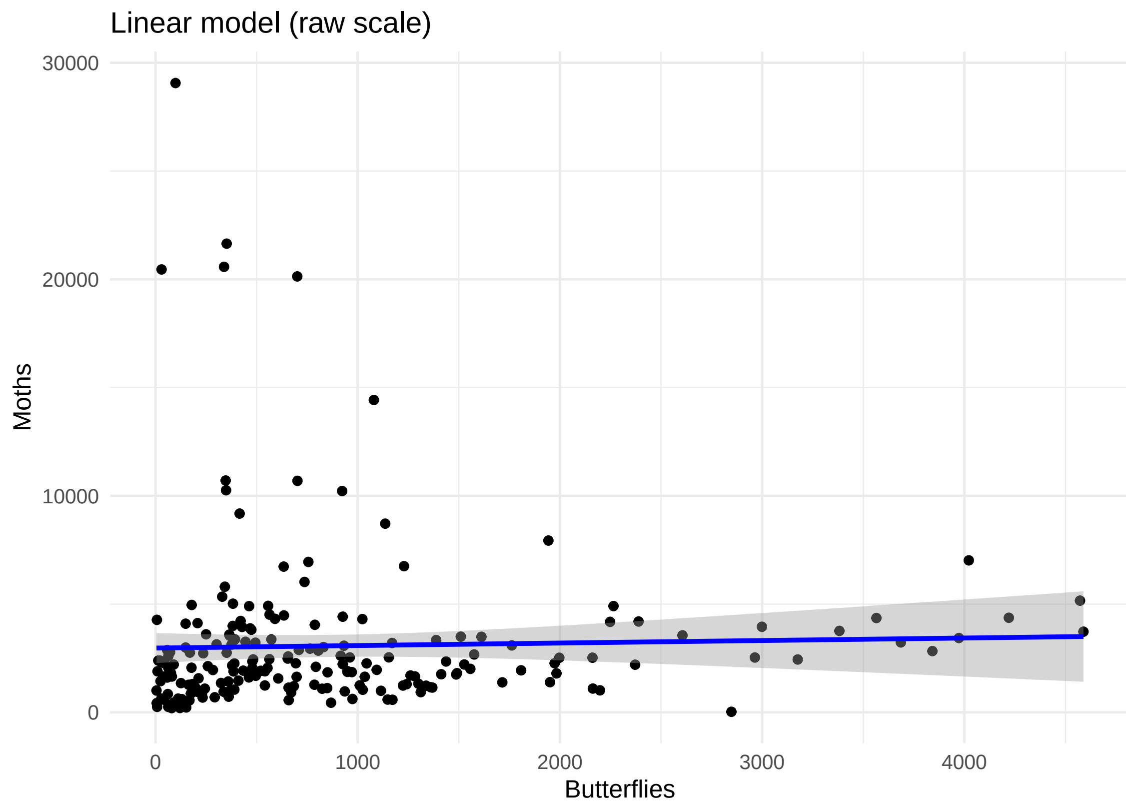
```r
# Load data
bf <- read.csv("ECN_IB1.csv", stringsAsFactors = FALSE)
mt <- read.csv("ECN_IM1.csv", stringsAsFactors = FALSE)

# Clean butterflies
bf$VALUE <- suppressWarnings(as.numeric(bf$VALUE))
bf$DATE  <- as.Date(bf$SDATE, format="%d-%b-%y")
bf$YEAR  <- year(bf$DATE)

bf_clean <- bf %>%
  filter(!is.na(VALUE), VALUE > 0, !is.na(YEAR))

# Clean moths
mt$VALUE <- suppressWarnings(as.numeric(mt$VALUE))
mt$DATE  <- as.Date(mt$SDATE, format="%d-%b-%y")
mt$YEAR  <- year(mt$DATE)

mt_clean <- mt %>%
  filter(!is.na(VALUE), VALUE > 0, !is.na(YEAR))

# Aggregate to annual totals
bf_agg <- bf_clean %>%
  group_by(SITECODE, YEAR) %>%
  summarise(total_butterflies = sum(VALUE), .groups="drop")

mt_agg <- mt_clean %>%
  group_by(SITECODE, YEAR) %>%
  summarise(total_moths = sum(VALUE), .groups="drop")

# Merge datasets
combined <- inner_join(bf_agg, mt_agg, by=c("SITECODE","YEAR"))

# Figure 1: Raw relationship
ggplot(combined, aes(total_butterflies, total_moths)) +
  geom_point() +
  labs(title="Butterflies vs Moths (raw)",
       x="Total annual butterfly counts",
       y="Total annual moth counts") +
  theme_minimal()
```



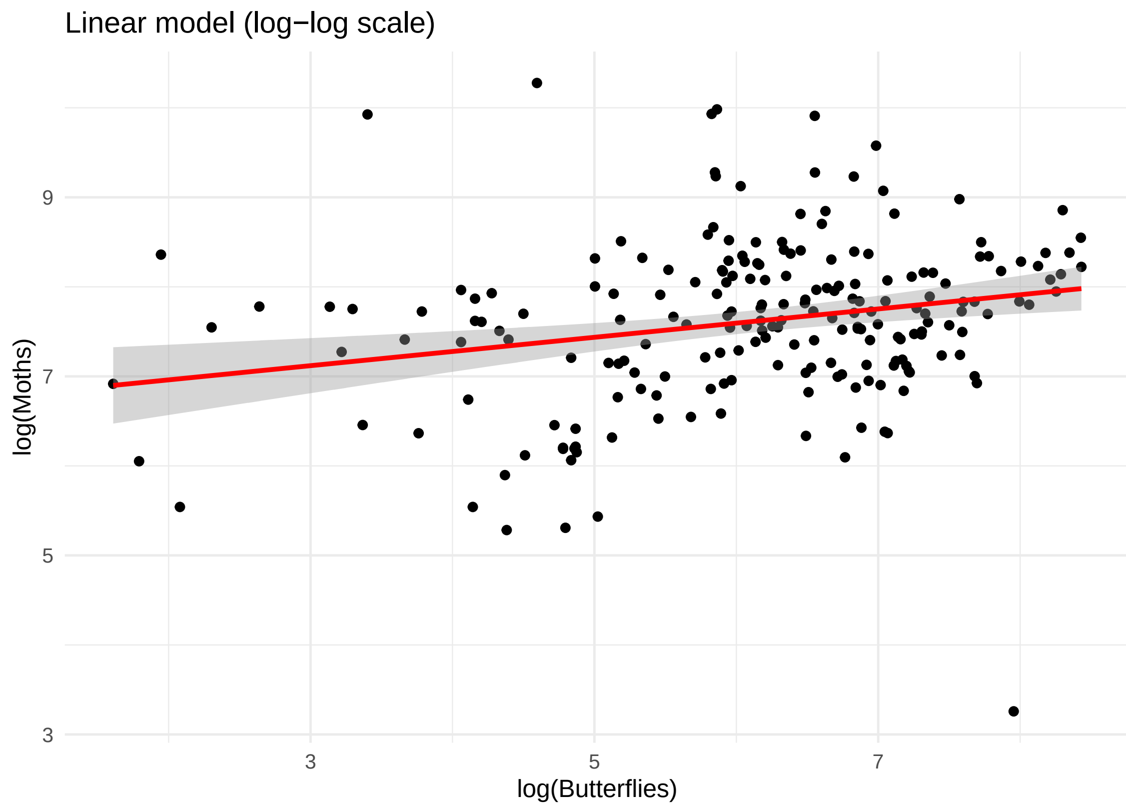

```
# Figure 2: Add grouping info
ggplot(combined,
       aes(total_butterflies, total_moths,
           colour=factor(YEAR), shape=SITECODE)) +
  geom_point(size=3, alpha=0.8) +
  labs(title="Butterflies vs Moths (by year & site)",
       x="Total annual butterfly counts",
       y="Total annual moth counts",
       colour="Year", shape="Site") +
  theme_minimal()
```



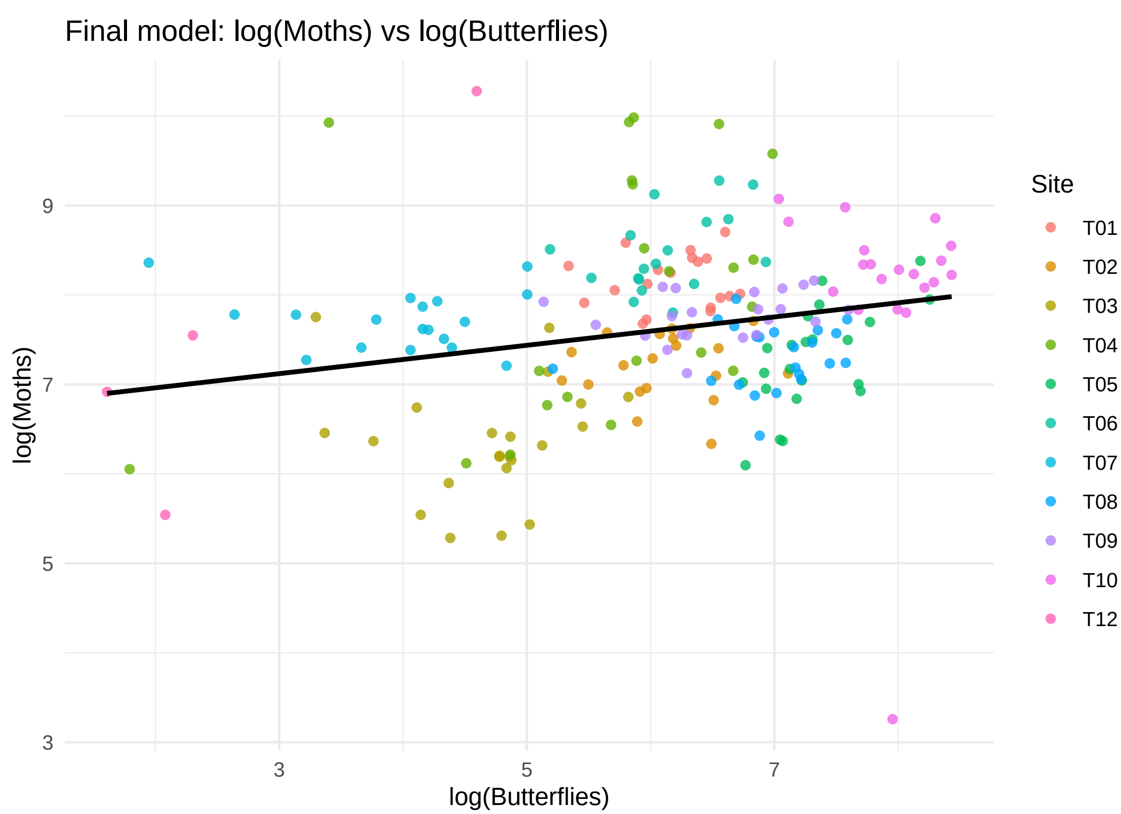

```r
# Linear regression models
model_raw <- lm(total_moths ~ total_butterflies, data=combined)

# Figure 3: Regression (raw scale)
ggplot(combined, aes(total_butterflies, total_moths)) +
  geom_point() +
  geom_smooth(method="lm", se=TRUE, colour="blue") +
  labs(title="Linear model (raw scale)",
       x="Butterflies",
       y="Moths") +
  theme_minimal()
```




```r
# Log transform (better fit)
combined <- combined %>%
  mutate(
    log_bf = log(total_butterflies),
    log_mt = log(total_moths)
  )

model_log <- lm(log_mt ~ log_bf, data=combined)

# Figure 4: Regression (log scale)
ggplot(combined, aes(log_bf, log_mt)) +
  geom_point() +
  geom_smooth(method="lm", se=TRUE, colour="red") +
  labs(title="Linear model (log-log scale)",
       x="log(Butterflies)",
       y="log(Moths)") +
  theme_minimal()
```




```
# Figure 5: Final model (clean)
ggplot(combined, aes(log_bf, log_mt)) +
  geom_point(aes(colour=SITECODE), alpha=0.8) +
  geom_smooth(method="lm", se=FALSE, colour="black", linewidth=1) +
  labs(title="Final model: log(Moths) vs log(Butterflies)",
       x="log(Butterflies)",
       y="log(Moths)",
       colour="Site") +
  theme_minimal()
```



```r
# Table 1: Coefficient summaries
raw_summary <- tidy(model_raw)
log_summary <- tidy(model_log)

raw_summary$model <- "Raw scale"
log_summary$model <- "Log-log scale"

combined_table <- bind_rows(raw_summary, log_summary) %>%
  select(model, term, estimate, std.error, statistic, p.value)

kable(combined_table, digits = 4,
      caption = "Linear regression coefficients (raw vs log-transformed)")
```

Table 1: Linear regression coefficients (raw vs log-transformed)

| model | term | estimate | std.error | statistic | p.value |
|---|---|---|---|---|---|
| Raw scale | (Intercept) | 2969.5437 | 351.9402 | 8.4376 | 0.0000 |
| Raw scale | total_butterflies | 0.1159 | 0.2757 | 0.4205 | 0.6746 |
| Log-log scale | (Intercept) | 6.6441 | 0.2881 | 23.0614 | 0.0000 |
| Log-log scale | log_bf | 0.1585 | 0.0461 | 3.4393 | 0.0007 |

```r
# Table 2: Model fit comparison
fit_table <- data.frame(
  Model = c("Raw scale", "Log-log scale"),
  R_squared = c(summary(model_raw)$r.squared,
                summary(model_log)$r.squared),
  Adj_R_squared = c(summary(model_raw)$adj.r.squared,
                    summary(model_log)$adj.r.squared),
  AIC = c(AIC(model_raw), AIC(model_log))
)

kable(fit_table, digits = 4,
      caption = "Model fit comparison (R², Adjusted R², AIC)")
```

Table 2: Model fit comparison ($R^2$, Adjusted $R^2$, AIC)

| Model | R_squared | Adj_R_squared | AIC |
|---|---|---|---|
| Raw scale | 0.0009 | -0.0040 | 4035.9438 |
| Log-log scale | 0.0541 | 0.0495 | 554.1964 |

The relationship between butterfly and moth counts was first explored using scatterplots. On the raw scale, there is a positive association, but the spread of the points increases as the counts get larger, which suggests heteroscedasticity. There are also a few extreme values, indicating potential outliers that could have a strong influence on the model fit.

A simple linear regression was initially fitted on the raw data. However, the assumptions of linear regression are not well satisfied here. In particular, the variance of the residuals is not constant, and the relationship does not appear strictly linear, showing some curvature.

To improve the model, both variables were log-transformed. After transformation, the relationship appears much more linear and the spread of the data is more consistent across the range of values. This suggests that the assumptions of linearity and homoscedasticity are better met.

The final model was therefore fitted using the log-transformed variables. This model provides a better fit to the data, and the slope can now be interpreted in terms of proportional changes (i.e an elasticity). Overall, the log transformation improves the suitability of the linear regression model for this type of ecological count data, which often shows skewness and non-constant variance.